BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 299 · Data mining and visualisation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is data mining?

Q2. What is an outlier?

Q3. What type of graph shows two numeric variables?

Q4. C# int[] scores={2,4,6,8}. Double elements at indexes 0 and 2; give the final values and total.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

data mining — Extracting patterns from datasets.

visualisation — Graphical representation of data.

correlation — Statistical association, not proof of cause.

outlier — Observation unusually far from others.

New learning

Identify the question — Data mining searches for useful patterns in a set of records; decide what question the analysis should answer.

Choose a suitable chart — Scatter charts show relationships between two numeric variables; bars compare categories; histograms show numeric distributions.

Interpret responsibly — An association does not prove that one variable causes another. Consider sample size, bias, missing data and outliers.

Worked example

For observations (1,2), (2,4), (3,6), describe the association without claiming causation.

1. Predict first — As x increases, y increases in this small sample, indicating positive association.

2. Explain the method — No causal claim is justified without additional evidence or experimental design.

We Do · Practise with guidance

For scores [40,42,45,90], select a suitable visual and explain a limitation of the mean.

```
Mean=(40+42+45+90)/4=54.25
Median=(42+45)/2=43.5
```

Predict first

A box plot or ordered dot plot shows the unusually high score 90 clearly.

Explain the method

The mean 54.25 is pulled upwards by the outlier; median 43.5 better describes the centre of the lower cluster.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Choose and justify a chart for (a) scores by category (b) minutes versus score (c) frequency by score interval. [6]

Question 2

For [40,42,45,90], calculate mean, median and identify one misleading interpretation. [6]

Question 3

Evaluate a school's plan to mine pupil browsing records to predict examination outcomes. [6]

Exit Ticket · Answer before checking

If two variables are correlated, the first must cause the second.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Identifying meaningful patterns in datasets.

Q2. Value unusually distant from the rest.

Q3. Scatter graph.

Q4. [4,4,12,8], total 28.

You Do pages · Independent questions

Q1. Bar chart categories [2]; scatter chart paired numeric measurements [2]; histogram of intervals [2].

Q2. Sum217 [1], mean54.25 [1]; ordered middle42,45 [1], median43.5 [1]; 90 is an outlier [1]; mean may hide clustering [1].

Q3. Purpose/legal necessity [1], minimise personal data [1], accuracy and confounding [1], fairness/bias [1], secure access [1], human interpretation rather than causal certainty [1].

Exit Ticket · Model answer

False. Confounding or coincidence can explain association.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.